

Sep 18.

Find the contours of a function numerically

Recall that we can express a function $f(x,y)$ numerically using a table:

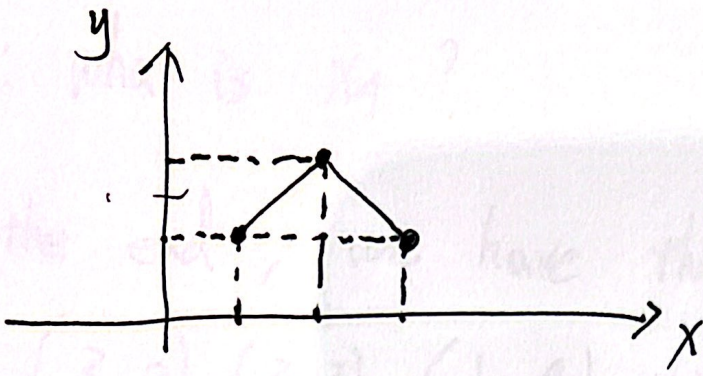
$y \backslash x$	0	2	4
0	12	8	4
2	10	6	2
4	8	4	0

Sketch an approximate contour ($f = 6$) within the sampled square.

Idea: assume that the function is linear between two adjacent sample points.

For example, if we know the value of a ^{one variable} function $h(x)$ at some sample points: $h(1)=1$ $h(2)=2$ $h(3)=1$

(1)



this may be a good approximation of the function h .

Now, we go back to our problem for $f(x, y)$.

First, we want to guess the point ~~x_0~~ x_0 such that ~~$f(x_0, 0) = 6$~~ $f(x_0, 0) = 6$.

Since $f(2, 0) = 8$, $f(4, 0) = 4$, if f is linear between these two points, then we have $f(3, 0) = \frac{f(2, 0) + f(4, 0)}{2} = 6$.

Thus, we guess $x_0 = 3$.

Then, we ~~find~~ want to find x_2 such that $f(x_2, 2) = 6$.

From the table, ~~$x_2 = 3$~~ $f(2, 2) = 6$. So $x_2 = 2$.

Then, we want to find x_4 such that $f(x_4, 4) = 6$.

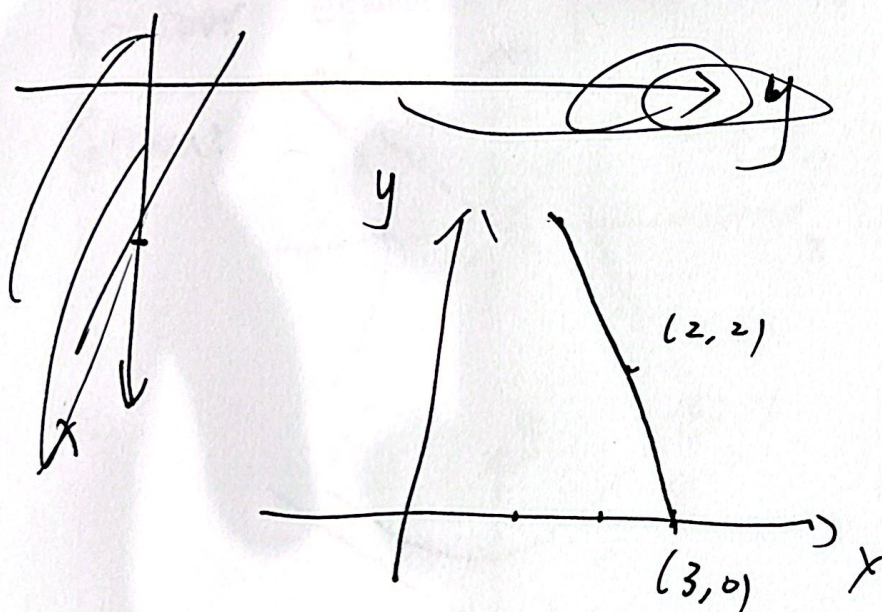
(2)

Q: what is x_4 ?

In the end, we have three possible points

$(3,0)$ $(2,2)$ $(1,4)$ which may lie in $\{f=6\}$

We just connect them to get an approximate contour $(f=6)$



Contour diagrams for some common functions

① $f(x,y) = x+2y$

Contour equation: $\{(x,y): x+2y=c\}$

Pattern: parallel lines.

②

② $f(x,y) = x^2 + y^2$

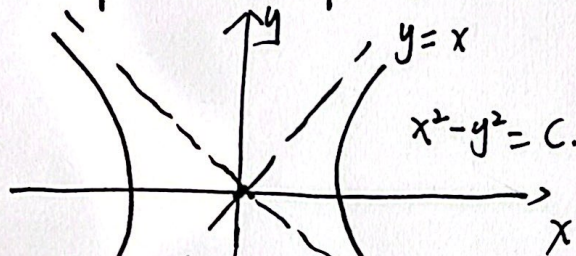
Contour equation: $\{(x,y): x^2 + y^2 = c\}$

Pattern: Circles for $c > 0$; a point for $c = 0$.

③. $f(x,y) = x^2 - y^2$.

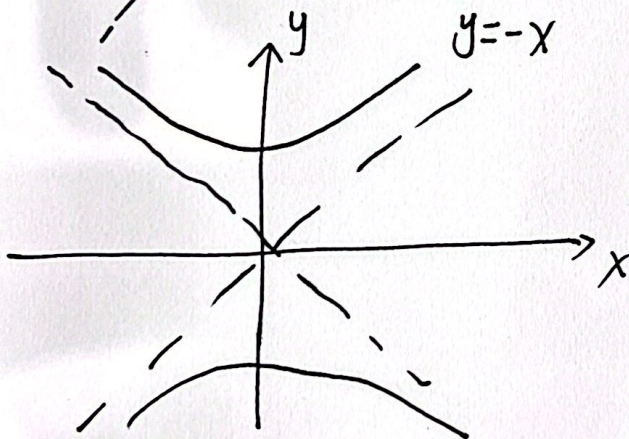
Contour equation: $\{(x,y): x^2 - y^2 = c\}$

Pattern:



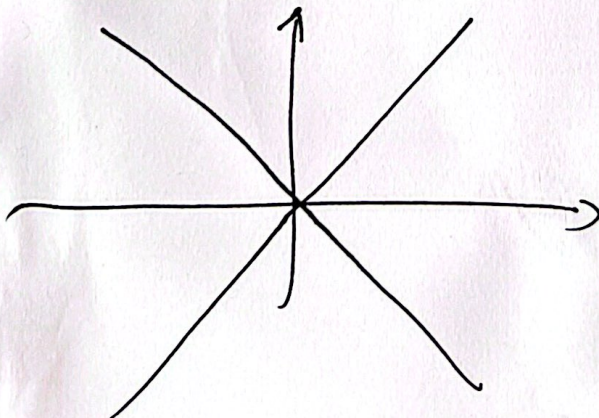
$c > 0$.

Hyperbolas



$c < 0$.

Hyperbolas



$c = 0$.

two intersecting lines.

(4)